



Performance Testing

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing
Type of Testing	Load Testing
Target Module / API	HDI Prediction
Test Environment	Local System
Test Date	Last test on 15/7/2026

Step 2: Test Scenarios



S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits HDI prediction request	1	30	Prediction result is displayed correctly.
2	Multiple prediction requests	10	60	Application responds successfully without errors.
3	Continuous prediction requests	20	120	Application remains stable and returns accurate predictions.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 seconds	1.2 seconds	Pass	Fast response for prediction requests.
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Maximum response time is within the limit.
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	System handled requests efficiently.
4	Error Rate	<1%	0%	Pass	No errors observed during testing.
5	CPU Utilization	<80%	45%	Pass	CPU usage remained stable.
6	Memory Utilization	<80%	52%	Pass	Memory usage was within acceptable limits.

Step 4: Observations & Analysis

Key Findings:

The HDI Prediction system performed efficiently during testing. The average responses time was below 2 seconds, CPU and memory usage remained within acceptable limits, and all prediction requests were processed successfully without failures.

Bottlenecks Identified:

No major performance bottlenecks were identified during testing. The application remained stable under the tested workload with no significant delays or errors.

Optimization Steps taken:

Input validation was implemented to prevent invalid data, unnecessary computations were minimized, and the trained machine learning model was loaded efficiently to improve prediction speed and overall application performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

The screenshot displays a web application titled "Human Development Index Predictor" under the heading "MACHINE LEARNING PROJECT". A green status badge in the top right corner indicates "ML Model Ready". The application's purpose is to "Estimate the Human Development Index using health, education and income indicators." The interface is divided into two main sections: "01 Development Indicators" and "02 HDI Analysis".

01 Development Indicators
Enter the required development values below.

Life Expectancy at Birth	Expected Years of Schooling
72 Years	13 Years
Mean Years of Schooling	Gross National Income Per Capita
8 Years	15000 USD

02 HDI Analysis
Machine learning based prediction.

Ready to Predict

Enter the development indicators and click the prediction button to generate the HDI score and

File

C:/Users/prems/OneDrive/Desktop/A%20Comprehensive%20Measure%20of%20WellBeing/3.%20Project%20Design%20Phase/index.html

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MACHINE LEARNING PROJECT

Human Development Index Predictor

Estimate the Human Development Index using health, education and income indicators.

ML Model Ready
01

Development Indicators

Enter the required development values below.

Life Expectancy at Birth

Example: 72

Years

Expected Years of Schooling

Example: 13

Years

Mean Years of Schooling

Example: 8

Years

Gross National Income Per Capita

Example: 15000

USD

Predict Human Development Index →

02

HDI Analysis